Distilling Segmenters From CNNs and Transformers for Remote Sensing Images' Semantic Segmentation

Zhe Dong, Guoming Gao, *Member, IEEE*, Tianzhu Liu, *Member, IEEE*, Yanfeng Gu, *Senior Member, IEEE*, and Xiangrong Zhang

***Abstract*— Semantic segmentation is a crucial task in remote sensing and has been predominantly performed using convolutional neural networks (CNNs) for the past decade. Recently, transformers with self-attention mechanisms have demonstrated superior performance compared with CNNs. However, due to the locality of CNN and the high computational complexity and massive data resource requirements of transformer, neither of them can be well applied in resource-constrained practical remote sensing scenarios. Motivated by the limitations of using either CNNs or transformers alone in the task of semantic segmentation of remote sensing images, a novel cross-model knowledge distillation (KD) framework, named distilling segmenters from CNNs and transformers (DSCTs), is proposed in this article to harness the complementary advantages of both the models. The framework uses a channel-weighted attention-guided feature distillation (CAFD) module to condense the feature from the teacher model and enhance the student model's focus on the teacher-focused regions. In addition, a target–nontarget KD (TNKD) module is proposed that decouples logit distillation into target and nontarget KD to guide the student model in learning the underlying representations and decision boundaries from the teacher model. By learning the complementary knowledge from the teacher, our proposed DSCT framework improves the student's segmentation performance without adding trainable parameters. Experiments on four available remote sensing datasets (ISPRS Potsdam, Vaihingen, GID, and LoveDA) indicate that the proposed DSCT outperforms the state-of-the-art KD methods and demonstrates its effectiveness and robustness.**

***Index Terms*— Convolutional neural networks (CNNs), knowledge distillation (KD), remote sensing images, semantic segmentation, transformer.**

I. Introduction

AS ONE of the essential tasks in remote sensing scenarios, semantic segmentation aims to categorize each pixel in a remote sensing image. This process is based on a comprehensive and dense semantic understanding and is considered a crucial step in the intelligent interpretation of remote sensing images. Its widespread application is evident in various fields, including hazard assessment [1], [2], urban planning [3], [4], agriculture [5], [6], environmental monitoring [7], [8], and so on.

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Zhe Dong, Guoming Gao, Tianzhu Liu, and Yanfeng Gu are with the School of Electronics and Information Engineering, Harbin Institute of Technology, Harbin 150001, China (e-mail: guyf@hit.edu.cn).

Xiangrong Zhang is with the Heilongjiang Institute of Technology, Harbin 150050, China.

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The rise of deep learning in the past decade has led to the widespread use of convolutional neural networks (CNNs) for automatic semantic segmentation. In particular, the introduction of fully convolutional networks (FCNs) has enabled end-to-end dense prediction. Subsequently, advances in encoder–decoder structures [9], [10], dilated convolutions [11], [12], multiscale feature fusion [13], [14], and attention mechanisms [15], [16] have been incorporated into the FCN-based models to achieve higher segmentation performance. The success of CNNs in semantic segmentation can be attributed to their inductive biases, such as locality and translation invariance, which enhance the learning efficiency of the network. However, the locality can also limit the ability of the CNN-based encoders to clearly learn long-range dependencies, which are crucial for scene understanding.

Inspired by the impressive performance of transformers [17] in natural language processing (NLP), researchers have sought to apply this technology to computer vision tasks. Unlike CNNs, transformers learn discriminative feature representations through self-attention mechanisms, enabling the modeling of rich interactions between pixels and the capture of global dependencies among tokens. The vision transformer (ViT) [18] was the first attempt to apply a pure transformer architecture for image recognition tasks and achieved competitive accuracy when pretrained on a large-scale dataset, such as ImageNet [19], and fine-tuned on downstream recognition benchmarks. Building on this structure, the segmentation transformer (SETR) [20] treats semantic segmentation as a sequence-to-sequence prediction task and uses global self-attention modeling for discriminative feature representation learning. Since then, transformer-based segmentation methods, such as Segmenter [21] and Segformer [22], have significantly outperformed previous state-of-the-art dense prediction approaches.

Although transformers have demonstrated superior performance in dense prediction tasks, such as semantic segmentation, compared with CNNs, they have not yet been widely adopted in remote sensing applications for two reasons. First, transformers lack the built-in inductive biases of CNNs and instead rely on data-driven learning, which requires large labeled datasets, making training more expensive. Second, the computational cost of the self-attention mechanisms in transformers increases quadratically with the size of the input, making it challenging to process high-resolution remote sensing images.

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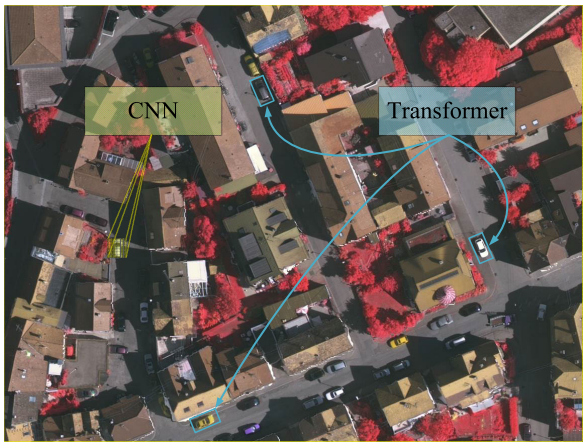


Fig. 1. Example of the strengths of CNNs and transformers illustrates that CNNs excel in capturing local spatial context while transformers are capable of modeling global interactions between pixels.

However, despite the advantages of CNNs and transformers in interpreting high-resolution remote sensing images, as illustrated in Fig. 1, their limitations hinder their effective application in resource-constrained practical remote sensing scenarios.

To address these challenges and leverage the complementary strengths of both the models, we propose a novel cross-model KD framework, called distilling segmenters from CNNs and transformers (DSCTs). In our DSCT framework, one of the CNN and transformer acts as the teacher, and the other acts as the student. Our DSCT framework aims to transfer comprehensive and complementary knowledge from the teacher network to the student network, thereby improving the segmentation performance of the student. Specifically, a channel-weighted attention-guided feature distillation (CAFD) module and a target–nontarget KD (TNKD) module are designed to transfer attention information and logit probability distributions in the proposed DSCT framework.

In summary, the main contributions of this article are given as follows.

1) A cross-model KD framework, called DSCTs, is proposed. This framework enables the CNN-based segmenters and the transformer-based segmenters to learn complementary knowledge through distillation, resulting in improved segmentation performance.
2) A CAFD module is designed to condense attention information in the feature map using channel weighting and pooling operations. The teacher model guides the student to focus on regions of interest, thereby improving the student's performance by mimicking the teacher's attention distribution.
3) In the proposed framework, a TNKD module that decouples the conventional logit distillation into target and nontarget KD is designed. It allows the student to learn the teacher's underlying representations and decision boundaries, enhancing the flexibility of knowledge transfer and reducing the suppression of nontarget knowledge compared with the conventional logit distillation.

The remainder of this article is organized as follows. In Section II, we discuss related works. Section III provides details on the proposed DSCT framework for cross-model KD. Section IV shows the experimental settings, ablation studies, and comparison results. Conclusions are illustrated in Section V.

II. Related Work

A. Vision Transformer

The transformer model [17], proposed by Google in 2017, has had a significant impact on the field of NLP. In recent years, there have been various innovative studies [23], [24], [25] that have applied transformers to various computer vision tasks across different fields, invigorating the field of computer vision and driving new innovations. Transformer's unique ability to learn long-distance dependencies and multimodal fusion has promoted its rapid development in the field of computer vision.

Dosovitskiy et al. [18] were the first to apply the original transformer to the task of image classification and proposed ViT, a structure that relies entirely on self-attention mechanisms. ViT has challenged the dominance of CNNs in computer vision, offering greater efficiency and scalability. However, ViT also has notable limitations. By slicing and flattening the image into a vector, it ignores the inherent properties and structure of the image. Furthermore, ViT requires substantial computing resources and may struggle to learn rich features with limited data. Aiming to address the limitations of ViT, Yuan et al. [26] introduced token-to-token ViT (T2TViT), which leverages a progressive tokenization mechanism to model both local structural information and global correlations in images. Jiang et al. [27] introduced a new training objective, token labeling, to fully leverage the potential of transformer in ImageNet classification. The CutMix [28] technology was also used to improve the performance and robustness of the model.

However, VIT and its variants only encode features at a single scale, which is not adequate for dense prediction tasks that require multiscale features. In recent years, there has been a growing interest in incorporating multiscale information into the transformer architecture to make it a general-purpose backbone for computer vision, similar to CNNs [29], [30], [31], [32], [33], [34], [35]. For example, the pyramid ViT (PVT) [29] proposed a hierarchical design for VIT, incorporating a spatial-reduction attention mechanism and a progressive shrinking pyramid. To mitigate the complexity of transformers, the Swin transformer [30] was proposed, which applied a shifted window strategy to allow cross-window information interaction with linear computational complexity. The Focal transformer [31] and CrossFormer [32], respectively, introduced focal self-attention and cross-scale embedding to capture both global and local relationships. The Multi-Scale Vision Longformer [33] applied local context in self-attention and used the efficient Longformer [34]. Besides, the CrossViT [35] encoded multiscale features through two branches and fuses them with a cross-attention module.

B. Semantic Segmentation Based on ViT

Although CNNs have an inductive bias that makes them more efficient and easier to train, they can also limit the ability

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to capture long-range dependencies, leading to performance degradation. The self-attention mechanisms in transformers help mitigate this issue by enabling the capture of global context, resulting in exceptional performance in semantic segmentation tasks [20], [21], [22].

The current state-of-the-art transformer-based semantic segmenters adopt either the ViT [18] or the Swin transformer [30] as the backbone for feature extraction. SETR [20] transformed semantic segmentation into a sequence-to-sequence prediction task, eliminating the need for models to learn local-to-global features by reducing resolution. SETR used ViT to decompose the image into fixed-size patches and then added the pixel vector and position embedding of each patch through linear transformations as the input to the encoder. The learned global features were then restored to the original image resolution by the decoder. Similarly, Segmenter [21] also used the ViT structure in the encoding stage, which divided the image into patches and performed linear mapping. In the decoding stage, the Segmenter applied either a point-wise linear decoder or a mask transformer decoder to obtain class labels from the output embeddings corresponding to image patches. Xie et al. [22] proposed SegFormer, a simple, efficient, and powerful framework for semantic segmentation, which combined the transformer with a lightweight multilayer perceptron (MLP) decoder. SegFormer used a hierarchical feature representation method to capture feature information at different scales and discarded the position embedding in ViT to avoid model performance degradation due to the differences in test and training image sizes. In addition, FASeg [36], Mask-former [37], PFD [38], and Mask2former [39] all achieved state-of-the-art performance on the semantic segmentation task using the Swin transformer as the backbone.

Although transformer-based semantic segmentation methods demonstrate superior segmentation performance compared with CNNs, their vast number of parameters and high computational cost present challenges in processing high-resolution images, particularly in remote sensing applications. In addition, due to the lack of inductive bias, transformers must be trained on large-scale datasets to maintain their competitive edge. As a result, efficiently and cost-effectively training transformers remains a significant challenge.

C. Knowledge Distillation

The concept of KD was first introduced in [40] as a method for transferring knowledge from a cumbersome teacher model to a smaller and more efficient student model. KD has been shown to enable student models to achieve substantial performance improvements while maintaining high efficiency. According to the transferred knowledge, the KD methods are mainly classified into feature-based, probability-based, and relation-based. The feature-based KD approaches [41], [42] use the refined features produced by the teacher model as a source of knowledge. The probability-based KD techniques [40], [43] use the class probabilities generated by the teacher model as soft labels to guide the training of the student model. The relation-based KD methods [44], [45], [46], [47] align the correlations or dependencies among instances between the student and teacher models.

Recently, researchers have made efforts to optimize the conventional KD methods to make them suitable for various computer vision tasks. FitNet [48] proposed not only to transfer the outputs but also the intermediate representations learned by the teacher. After that, RKD [44] was designed to transfer mutual relationships of data examples. OFD [41] discovered the effectiveness of the location of activation function and proposed a new loss function to improve the performance of feature distillation. Review KD [49] applied the review mechanism connected by cross-level paths between teacher–student networks. DKD [50] decoupled the conventional KD method to verify the potential of logit distillation. Liu et al. [51], [52] proposed the structured KD (SKDS) method, which transfers structure knowledge from a teacher model to a student model through adversarial training and pairwise relationships. Xie et al. [53] proposed a teacher–student learning framework that transfers both zero-order and first-order knowledge by aligning local similarity maps. He et al. [54] aimed to minimize feature discrepancy by transferring non-local pairwise affinity maps using an additional autoencoder. An intraclass feature variation distillation (IFVD) was introduced in [55] to transfer the intraclass feature variation (IFV) of the teacher model to the student model. Shu et al. [56] designed a channelwise KD method to guide the student model in aligning the activations of the corresponding channels in the teacher model. Cross-image relation KD (CIRKD) [57] focused on transferring cross-image global pixel correlations rather than structured information.

III. Proposed Framework

A. Overview

The purpose of our framework is quite straightforward: Improving segmentation performance through transferring complementary knowledge between CNNs and transformers. Semantic segmentation is an intensive pixel-level prediction task that assigns each pixel in an image a specific category. Given an input image $\mathcal{I} \in \mathbb{R}^{H\times W\times D}$, the feature extractor of a segmenter first generates a feature map $\mathcal{F} \in \mathbb{R}^{\bar{H}\times \bar{W}\times C}$, where H, W, D and $\bar{H}$, $\bar{W}$, C denote the height, width, and the number of channels of the input image and feature map, respectively. The decode head further transforms the feature map $\mathcal{F}$ into a categorical logit map $\mathcal{Z} \in \mathbb{R}^{H\times W\times N}$, with N being the number of prediction classes. The conventional segmentation task loss is to train each pixel of $\mathcal{Z}$ with the corresponding ground-truth label $\mathcal{Y}$ using cross-entropy loss

$$\mathcal{L}_{\text{seg}} = \frac{1}{H\times W}\sum_{h=1}^{H}\sum_{w=1}^{W}\text{CE}\left(\sigma\left(\mathcal{Z}_{(h,w)}\right), \mathcal{Y}_{(h,w)}\right) \tag{1}$$

where $\text{CE}(\cdot)$ represents the cross-entropy loss, σ is the softmax activation, and $\mathcal{Z}_{(h,w)}$ and $\mathcal{Y}_{(h,w)}$ denote the value of the (h, w)-th pixel of $\mathcal{Z}$ and $\mathcal{Y}$, respectively.

The existing KD methods usually use a pixelwise alignment among class probabilities between a cumbersome teacher model t and a compact student model s based on the task

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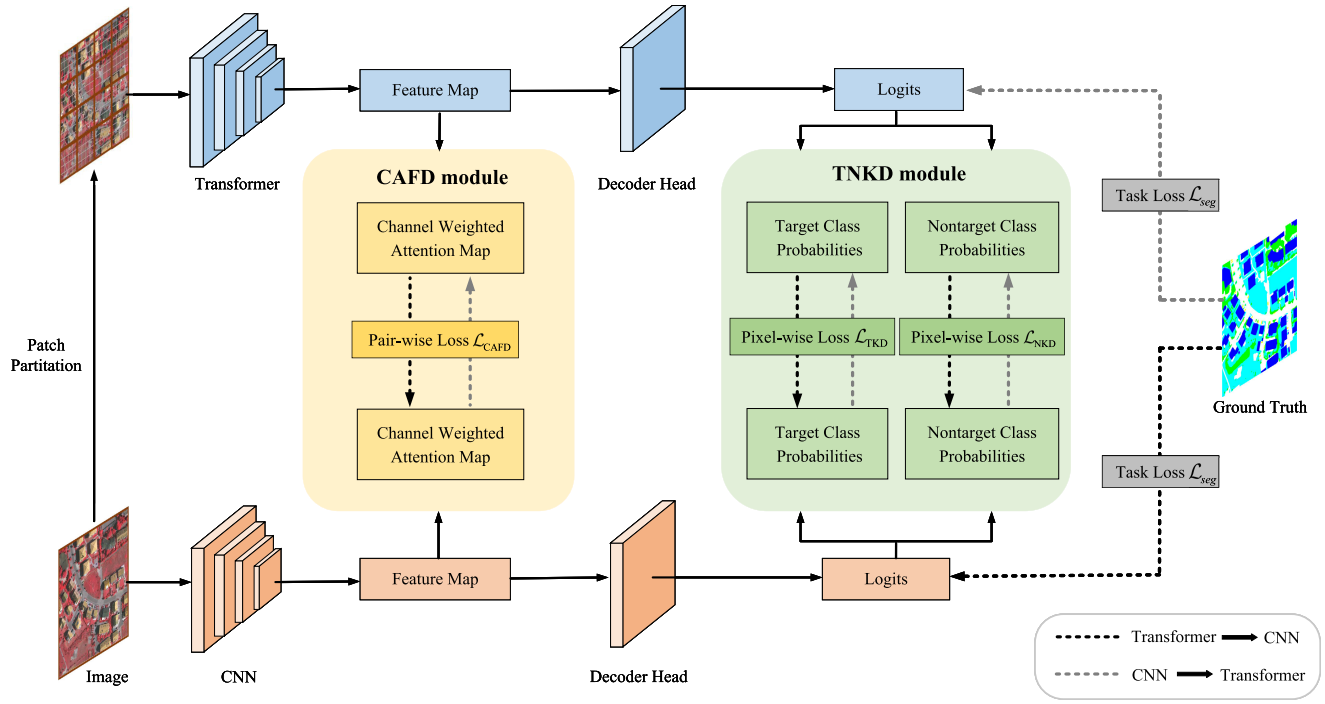


Fig. 2. Illustration of the proposed DSCT. For both CNN → transformer and transformer → CNN distillation processes, we use a trained teacher model that remains frozen during the training of the student. The object of the training is to minimize both the task loss $\mathcal{L}_{\text{seg}}$ and the distillation loss $\mathcal{L}_{\text{CAFD}}$ and $\mathcal{L}_{\text{TNKD}}$.

loss, which can be formulated as

$$\mathcal{L}_{\text{seg}} + \alpha \cdot \frac{1}{H \times W} \sum_{h=1}^{H} \sum_{w=1}^{W} \text{KL}\left(\sigma\left(\frac{\mathcal{Z}^t_{(h,w)}}{\tau}\right) \middle\| \sigma\left(\frac{\mathcal{Z}^s_{(h,w)}}{\tau}\right)\right) \tag{2}$$

where $\sigma(Z^t_{(h,w)}/\tau)$ and $\sigma(Z^s_{(h,w)}/\tau)$ denote the soft class probabilities of the (h, w)th pixel of t and s, respectively, α is a hyperparameter to balance the loss item, $\text{KL}(\cdot)$ represents the Kullback–Leibler divergence [40] between two probabilities, and τ is the temperature to soften the output. While logits seem to be a more efficient source of knowledge, CNNs and transformers fail to learn feature representations and spatial dependencies from each other [58]. Besides, the majority of KD methods rely on two homologous models of different sizes, ignoring the potential of cross-model distillation.

To address these limitations, we propose a bidirectional KD framework to learn complementary knowledge between CNNs and transformers, called DSCTs, as illustrated in Fig. 2. The semantic segmenter consists of a backbone architecture and a decoder head. The backbone, which can be either a CNN or a transformer, takes an image as input and extracts its feature maps, and the decoder head then upsamples the feature maps to generate class probability values. In general, we choose one of the CNNs or transformers as the teacher model, while the other serves as the student model. Before KD, the teacher model is first pretrained on an annotated dataset. We then freeze the parameters of the teacher model and the student model is trained with ground truth and the knowledge extracted from the teacher model.

B. Channel-Weighted Attention-Guided Feature Distillation

Attention transfer is a special form of feature distillation, which aims to let the student model concentrate more on the areas that are of interest to the teacher and imitate the teacher's attention allocation. For the conventional attention transfer methods [42], [59], given a feature map $\mathcal{F} \in \mathbb{R}^{\tilde{H} \times \tilde{W} \times C}$, an attention mapping $\mathcal{M}$ takes $\mathcal{F}$ as input and outputs a 2-D spatial attention map $\mathcal{F}_A$, which can be written as

$$\mathcal{M} : \mathcal{F} \in \mathbb{R}^{\tilde{H} \times \tilde{W} \times C} \rightarrow \mathcal{F}_A \in \mathbb{R}^{\tilde{H} \times \tilde{W}}. \tag{3}$$

The attention transfer loss $\mathcal{L}_{\text{AT}}$ can be expressed as

$$\mathcal{L}_{\text{AT}} = \left\| \frac{\mathcal{F}^t_A}{\left\|\mathcal{F}^t_A\right\|_2} - \frac{\mathcal{F}^s_A}{\left\|\mathcal{F}^s_A\right\|_2} \right\|_2 \tag{4}$$

where $\mathcal{F}^t_A$ and $\mathcal{F}^s_A$ denote the spatial attention maps of the teacher and student models, respectively.

However, equal contributions from each layer of the feature map $\mathcal{F}$ during the attention transfer process can lead to the inclusion of ineffective and redundant information, thereby diluting the accurate and meaningful information from the teacher model.

To extract practical and accurate knowledge with channel information from the feature map of the teacher model, we proposed a CAFD module to assign different channel ratios for attention mapping operation. As demonstrated in Fig. 3, we introduce the channel attention mechanism [60] to measure the channel weight ratio, and then assign the corresponding weight to each layer of the feature map through a scale operation. This not only preserves spatial dependencies but also incorporates channel information in the attention transfer.

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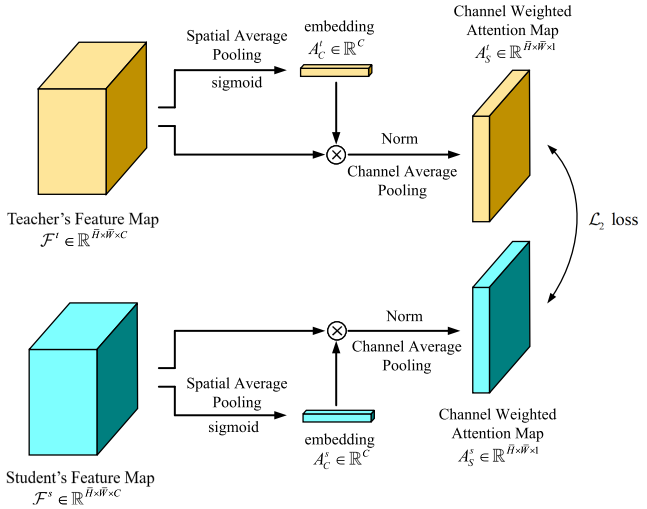


Fig. 3. Illustration of the proposed CAFD module for computing the pairwise distillation loss $\mathcal{L}_{\text{CAFD}}$.

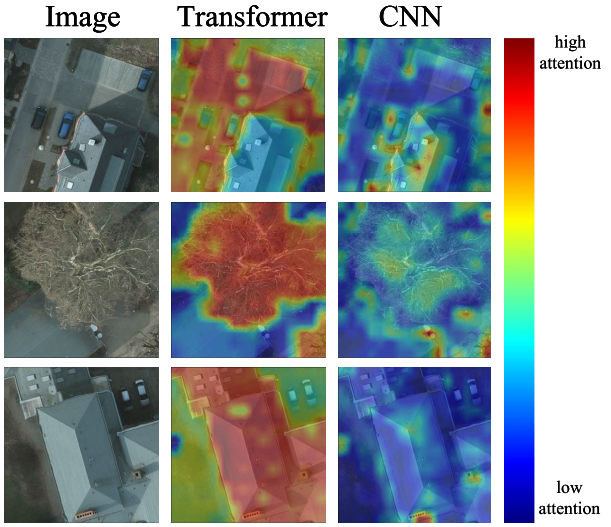


Fig. 4. Visualization of the spatial attention maps output by transformers and CNNs. The images in the first column represent the input remote sensing images; the ones in the second and third columns represent the spatial attention maps output by transformer (Swin transformer) and CNN (ResNet).

For the feature map $\mathcal{F}$ output by the feature extractor, we apply a spatial average pooling operation to squeeze $\mathcal{F}$ into a feature embedding of length C and then introduce the sigmoid activation to restrict it to (0, 1) to obtain the channel attention embedding $A_C \in \mathbb{R}^C$, which can be expressed as

$$A_C = \delta(P_s(\mathcal{F})) = \delta\left(\frac{1}{\bar{H}\times\bar{W}}\sum_{i=1}^{\bar{H}}\sum_{j=1}^{\bar{W}}\mathcal{F}_{(i,j,\cdot)}\right) \tag{5}$$

where $P_s(\cdot)$ represents the spatial average pooling operation, δ is the sigmoid activation, and i, j denote the ith, jth slice of $\mathcal{F}$ in height and width dimensions. A_C indicates the weight of each channel in the feature map.

Through channelwise multiplication between the channel attention embedding A_C and the feature map $\mathcal{F}$, we can get the channel-weighted feature map $\widetilde{\mathcal{F}}$. Then we apply normalization to remove the deleterious magnitude of $\widetilde{\mathcal{F}}$

$$\widetilde{\mathcal{F}}_N = \text{Norm}\left(\widetilde{\mathcal{F}}\right) = \text{Norm}(A_C \otimes \mathcal{F}) \tag{6}$$

where $\widetilde{\mathcal{F}}_N$ is the normalized $\widetilde{\mathcal{F}}$, $\otimes$ refers to the channelwise multiplication operation, and Norm represents normalization. In this article, we select the instance normalization along the $\bar{H}\times\bar{W}$ dimension as the normalization.

Different from the maximum or summation operation in [42], we introduce the channel average pooling P_c as the attention mapping function to obtain the spatial attention map $A_S \in \mathbb{R}^{\bar{H}\times\bar{W}\times 1}$

$$A_S = P_c\left(\widetilde{\mathcal{F}}_N\right) = \frac{1}{C}\sum_{k=1}^{C}\widetilde{\mathcal{F}}_{N(\cdot,\cdot,k)}. \tag{7}$$

Similarly, k denotes the kth slice of $\widetilde{\mathcal{F}}$ in the channel dimension.

To allow the student to learn effective spatial dependencies from the teacher's spatial attention map, we minimize the $\mathcal{L}_2$ distance between the spatial attention maps of the teacher and student, which can be expressed as

$$\begin{aligned}\mathcal{L}_{\text{CAFD}} &= \mathcal{L}_2\left(A_S^t, A_S^s\right)\\ &= \frac{1}{\bar{H}\times\bar{W}}\left\|\left(A_S^t, A_S^s\right)\right\|_2^2\end{aligned} \tag{8}$$

where A_S^t and A_S^s denote the spatial attention map of the teacher and student, respectively.

To illustrate the contrast in spatial attentions between CNNs and transformers, we visualize some spatial attention maps for both, which can be viewed in Fig. 4. It is evident that CNNs and transformers exhibit vastly different activation responses and spatial dependencies. As a result, the proposed CAFD module can aid the student in mimicking the attention maps of the teacher model, thereby enhancing the student's segmentation performance by providing information on the areas where the teacher focuses its attention.

C. Target–Nontarget KD

Compared with feature-based distillation, the performance of logit distillation is inferior. However, logit distillation has the potential to achieve comparable performance to feature distillation because logits contain more high-level semantic information. Recent study [50] has indicated that the traditional KD loss function is tightly coupled and can be reformulated as

$$\mathcal{L}_{\text{KD}} = \text{KL}\left(\mathcal{Z}_T^t, \mathcal{Z}_T^s\right) + (1-p)\text{KL}\left(\left(\mathcal{Z}_N^t, \mathcal{Z}_N^s\right)\right) \tag{9}$$

where $\mathcal{Z}_T^t, \mathcal{Z}_T^s \in \mathbb{R}^{H\times W\times 1}$ denote the soft target class probabilities of the teacher and student, $\mathcal{Z}_N^t, \mathcal{Z}_N^s \in \mathbb{R}^{H\times W\times(N-1)}$ denote the soft nontarget class probabilities of the teacher and student, and p is the prediction confidence of the target class. A higher confidence value p suggests that the knowledge imparted by the teacher model is more trustworthy; however, it also decreases the representation of knowledge pertaining to nontarget classes.

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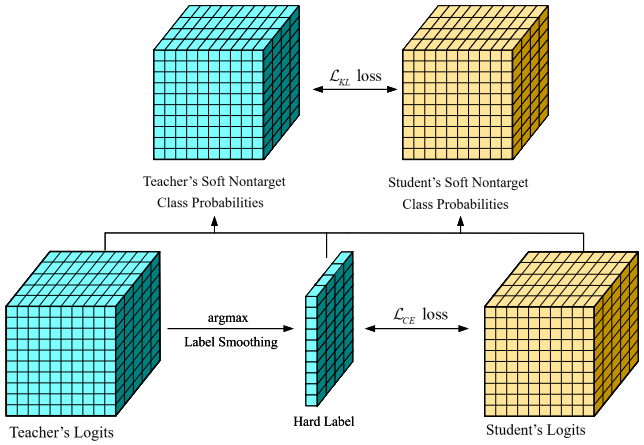


Fig. 5. Illustration of the proposed TNKD module for computing the pixel-wise distillation loss $\mathcal{L}_{\text{TNKD}}$. The TNKD module decouples the conventional KD into a weighted sum of two parts, i.e., TKD and NKD. For TKD, the teacher's hard label is used as the target to guide student training. While NKD transfers nontarget knowledge by aligning the probability distributions of the teacher and students' nontarget categories.

Inspired by this, we proposed a TNKD module, which comprehensively considers the effect of the target class KD and nontarget class KD. To enhance the flexibility for knowledge transfer and unsuppress the effect of nontarget knowledge, we decouple the categorical logits into two independent components: target KD (TKD) and nontarget KD (NKD), which can be seen in Fig. 5. TKD aims to transfer the predictive knowledge of the teacher model to the student model, thereby helping the student model improve its decision-making ability, while NKD transfers auxiliary knowledge that is not directly related to the target task, including background information and category correlation, so that the student model can better understand the characteristics, relationships, and representations of data.

For the logits $\mathcal{Z}^t \in \mathbb{R}^{H\times W\times N}$ output from the teacher model, we apply the arg max function to obtain the target class index among N channels. In this way, the probability distribution of $\mathcal{Z}^t$ is transformed into the hard label, which plays the same role as the ground truth $\mathcal{Y}$. Then a label smoothing [61] regularization strategy is introduced to reduce the weight of the target categories when calculating the loss function, thus preventing overfitting. The target KD loss $\mathcal{L}_{\text{TKD}}$ can be written as follows:

$$\mathcal{L}_{\text{TKD}} = \mathcal{L}_{\text{CE}}\left(\mathcal{Z}^s, \zeta\left(\arg\max\left(\mathcal{Z}^t\right)\right)\right) \tag{10}$$

where ζ denotes the label smoothing operation.

For the nontarget KD, we apply the Kullback–Leibler divergence to align per-pixel nontarget class probabilities produced from teacher and student models, which can be formulated as

$$\begin{aligned}\mathcal{L}_{\text{NKD}} &= \text{KL}\left(\mathcal{Z}_N^t, \mathcal{Z}_N^s\right) \\ &= \frac{\tau}{(N-1)}\sum_{n=1}^{N-1}\sum_{m=1}^{W\cdot H}\sigma\left(\mathcal{Z}_{N,(n,m)}^t\right)\cdot\log\left[\frac{\sigma\left(\mathcal{Z}_{N,(n,m)}^t\right)}{\sigma\left(\mathcal{Z}_{N,(n,m)}^s\right)}\right]\end{aligned} \tag{11}$$

where $\mathcal{Z}_{N,(n,m)}^t$ and $\mathcal{Z}_{N,(n,m)}^s$ present the soft class probabilities of $\mathcal{Z}_N^t$ and $\mathcal{Z}_N^s$ at (n, m) position, respectively.

Our proposed TNKD can be expressed as a combination of TKD and NKD, and we introduce two hyperparameters λ_1 and λ_2 to balance the weight of them. The loss function of TNKD can be written as follows:

$$\mathcal{L}_{\text{TNKD}} = \lambda_1 \cdot \mathcal{L}_{\text{TKD}} + \lambda_2 \cdot \mathcal{L}_{\text{NKD}}. \tag{12}$$

In our TNKD, it is allowed to change each term's weight to balance the importance of TKD and NKD and distill effective knowledge from logits in a flexible and efficient manner.

Algorithm 1 Distilling Segmenters From CNNs and Transformers (DSCTs)

Input:
$D = \{\mathcal{I}, \mathcal{Y}\}$ is data for training, $\mathcal{I}$ and $\mathcal{Y}$ are the images and the corresponding ground truths, respectively;
T, θ_t are the teacher model and its parameters;
Output:
S, θ_s are the student model and its parameters;
Load the pretrained weights of the teacher model T and initialize the parameters of the student model S.
for each batch $D_b = \{\mathcal{I}_b, \mathcal{Y}_b\}$ **do**
  Use the teacher model T on the batch D_b:
    $\left\{\mathcal{F}^t, \mathcal{Z}^t\right\} \leftarrow T(\mathcal{I}_b, \theta_t)$;
  Use the student model S on the batch D_b:
    $\{\mathcal{F}^s, \mathcal{Z}^s\} \leftarrow S(\mathcal{I}_b, \theta_s)$;
  Calculate the channel-weighted attention-guided feature distillation loss $\mathcal{L}_{\text{CAFD}}$ (8);
  Calculate the target–nontarget knowledge distillation loss $\mathcal{L}_{\text{TNKD}}$ (12);
  Calculate the segmentation task loss $\mathcal{L}_{\text{seg}}$ (1);
  Calculate the total loss $\mathcal{L}_{total}$ (13);
  Update the parameters θ_s of the student model S:
    $\theta_s = \text{argmin}_{\theta_s}\, \mathcal{L}_{total}$;
end for

D. Optimization

Our proposed cross-model KD framework comprises a CNN-based segmenter and a transformer-based segmenter, with one serving as the teacher and the other as the student. During the training process, the segmentation task loss $\mathcal{L}_{\text{seg}}$ and our designed distillation loss $\mathcal{L}_{\text{distill}}$ are used to train the student model, while the parameters of the teacher model are kept fixed. The distillation loss is composed of two parts: $\mathcal{L}_{\text{CAFD}}$ and $\mathcal{L}_{\text{TNKD}}$. Thus, the total loss can be defined as follows, taking into account both the components of the distillation loss:

$$\begin{aligned}\mathcal{L}_{\text{total}} &= \mathcal{L}_{\text{seg}} + \mathcal{L}_{\text{distill}} \\ &= \mathcal{L}_{\text{seg}} + \beta\mathcal{L}_{\text{CAFD}} + \mathcal{L}_{\text{TNKD}} \\ &= \mathcal{L}_{\text{seg}} + \beta\mathcal{L}_{\text{CAFD}} + \lambda_1\mathcal{L}_{\text{TKD}} + \lambda_2\mathcal{L}_{\text{NKD}}.\end{aligned} \tag{13}$$

The whole training process of our DSCT framework is illustrated in Algorithm 1.

IV. Experiments

In this section, extensive experiments are conducted to evaluate the effectiveness of our proposed method.

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A. Dataset

We evaluate our proposed method on four publicly available remote sensing image semantic segmentation datasets: the International Society for Photogrammetry and Remote Sensing (ISPRS) 2-D Semantic Labeling Challenge Potsdam and Vaihingen datasets,[1] the Gaofen Image Dataset (GID) [62], and LoveDA dataset [63].

1) Potsdam: The Potsdam dataset consists of 38 true orthophoto (TOP) tiles and their corresponding digital surface models (DSM); 24 images are used for training and the remaining 14 images for testing. All the tiles have four bands, i.e., red (R), blue (B), green (G), and near-infrared (NIR). Each tile has a spatial size of 6000 × 6000 pixels, and the ground sampling distance (GSD) is 5 cm. The dataset is annotated with six categories for semantic segmentation: impervious surfaces, building, low vegetation, tree, car, and clutter/background.

2) Vaihingen: The Vaihingen dataset is composed of 33 TOP tiles with an average size of 2494 × 2064, 16 of which are fully annotated. Among the 16 annotated tiles, 11 images are used as the training set, and the other images are adopted for testing. Unlike the Potsdam dataset, each TOP in the Vaihingen dataset contains three bands (NIR, R, and G) and one DSM band, and the GSD is 9 cm. The Vaihingen dataset has the same conventional ground object types as the Potsdam dataset.

3) GID: The GID dataset consists of 10 high-resolution tiles captured by the Gaofen-2 Satellite. Each tile has a spatial size of 7200 × 6800 pixels and contains R-G-B spectral bands. Besides, each tile covers a geographic region of 506 km^2 and is labeled with 15 classes of land use categories. Since the division of training and testing sets is not provided, eight out of the 10 tiles are chosen for training, are the remaining two are used for testing.

4) LoveDA: The LoveDA dataset contains 5987 high spatial resolution (0.3 m) remote sensing images of 1024 × 1024 pixels and RGB spectral bands. It is separated into two parts: urban and rural, and each part consists of the training set, validation set, and test set. The LoveDA dataset contains seven categories: background, building, road, water, barren, forest, and agriculture. Since the annotations of the test set are not available, we apply the validation test for testing.

B. Implement Details

1) Teacher and Student Models: A segmenter typically has two components: a backbone that extracts features and a head that decodes the features back to the size of the original image. Given the considerations of efficiency and compatibility, we choose UPerNet [64] as the decoder head for both the CNN-based and the transformer-based segmenters. The backbone of the CNN-based segmenter is ResNet [65], while the Swin transformer [30] serves as the backbone of the transformer-based segmenter. In the following, we present simplified representations of the CNN-based and transformer-based segmenters using ResNet and Swin, respectively. When one of these models is designated as the teacher, the other serves as the student.

2) Training Setup: Our network is implemented using the PyTorch framework. We use the adaptive moment estimation (AdamW) optimizer [66] to train all our models, with an initial learning rate of 3e-5 and a weight decay of 0.01. To dynamically adjust the learning rate, we use the "Poly" decay strategy [67]. For the setting of hyperparameters, following previous works [52], [55], we set the temperature τ to 1.0. Besides, the weight coefficients β, λ_1, and λ_2 are set to 5.0, 1.0, and 8.0, respectively. During both training and inference, we crop images into nonoverlapping patches of size 512 × 512 pixels. Standard data augmentation strategies, including random resizing, cropping, flipping, and photometric distortion, are applied during model training. We train our models on two NVIDIA Tesla V100 GPUs with 32-GB memory, with a batch size of 8 and a maximum number of epochs set to 80, for all the training runs.

3) Evaluation Metrics: For all the datasets, we evaluate the results of semantic segmentation using three metrics: overall accuracy (OA), F_1 score, and mean intersection-over-union (mIoU) of pixels. OA represents the probability that all the pixels are correctly classified and can be expressed as

$$\mathrm{OA} = \frac{(\mathrm{TP} + \mathrm{TN})}{(\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN})} \tag{14}$$

where TP, FP, TN, and FN stand for true positive, false positive, true negative, and false negative, respectively. These four indicators represent the situation when different pixels are predicted to different categories.

The F_1 score is the harmonic average of precision and recall of a particular class, and it is calculated using the following formula:

$$\begin{aligned} \mathrm{precision} &= \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \\ \mathrm{recall} &= \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \\ F_1 &= 2 \times \frac{\mathrm{precision} \times \mathrm{recall}}{\mathrm{precision} + \mathrm{recall}}. \end{aligned} \tag{15}$$

The mIoU represents the average intersection-over-union over all the categories, which is expressed as follows:

$$\mathrm{mIoU} = \frac{1}{k} \sum_{i=0}^{k} \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}. \tag{16}$$

Besides, the network parameters are applied to measure the model size.

C. Ablation Study

In this section, we demonstrate the effectiveness of the DSCT and its submodules through comprehensive ablation studies conducted on the LoveDA urban dataset and Potsdam dataset.

1) Effectiveness of DSCT: To verify the effectiveness of cross-model distillation, we apply the proposed DSCT as the KD method and evaluate a set of (teacher, student) pairs. Specifically, we consider the following models: ResNet-18 and ResNet-101 (CNN models) and Swin-T and Swin-B

[1] https://www2.isprs.org/commissions/comm2/wg4/benchmark/semantic-labeling

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TABLE I

MIOU OF THE STUDENT MODEL OF VARIOUS (TEACHER, STUDENT) PAIRS ON THE LOVEDA URBAN VALIDATION SET. "PARAMS" DENOTES THE NETWORK PARAMETERS AND "W/O DIS" REPRESENTS THE STUDENT MODEL IS TRAINED WITHOUT DISTILLATION. "(+)" AND "(−)" INDICATE IMPROVEMENT AND DECREASE IN PERFORMANCE, RESPECTIVELY

Student	Params (M)	Teacher				
		w/o dis	ResNet-18	ResNet-101	Swin-T	Swin-B
ResNet-18	41.50	48.12	48.16 (+0.04)	49.57 (+1.45)	50.02 (+1.90)	**50.89 (+2.77)**
ResNet-101	85.51	51.87	51.83 (-0.04)	51.90 (+0.03)	52.19 (+0.32)	**52.74 (+0.87)**
Swin-T	59.94	50.32	50.86 (+0.54)	**51.92 (+1.60)**	50.41 (+0.09)	51.26 (+0.94)
Swin-B	121.28	53.04	53.19 (+0.15)	**53.35 (+0.31)**	52.93 (-0.11)	53.11 (+0.07)

TABLE II

EFFECTIVENESS OF THE PROPOSED DSCT DISTILLATION FRAMEWORK AND ITS SUBMODULES CAFD AND TNKD

Method	CAFD	TNKD	mIoU(%)
T:Swin-B			84.22
S: ResNet-18	n/a	n/a	82.12
	✓	✗	83.04
	✗	✓	83.19
	✓	✓	**83.73**
T: ResNet-101			83.77
S: Swin-T	n/a	n/a	83.63
	✓	✗	84.03
	✗	✓	84.12
	✓	✓	**84.29**

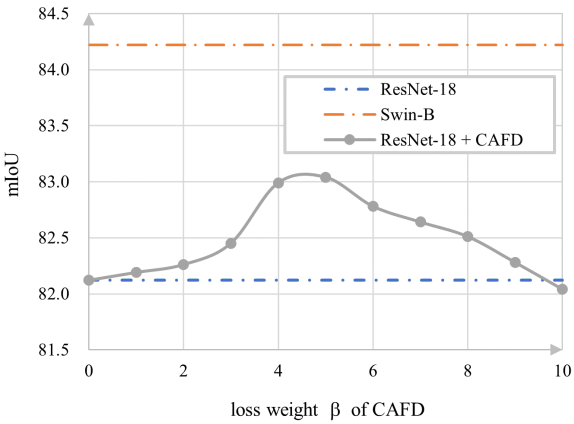


Fig. 6. Impact of different loss weights β of the CAFD module.

(transformer models). In this experiment, both the teacher and student models are trained on the LoveDA urban train set and then evaluated on the LoveDA urban validation set. We report the segmentation performance of student models trained under the guidance of different teacher models. As shown in Table I, we observe that cumbersome teacher models can lead to performance improvements for smaller student models. For instance, when ResNet-101 is selected as the teacher, it leads to 1.45% and 1.60% improvements in mIoU for students ResNet-18 and Swin-T, respectively. And the teacher Swin-B brings 2.77% and 0.94% mIoU improvements for students ResNet-18 and Swin-T, respectively. Besides, we also find that students consistently perform better when their teachers have different architectures. Moreover, it can be seen that a smaller model in different architecture can also be a good teacher. When ResNet-18 is viewed as the teacher, the student Swin-B has a performance enhancement of 0.15% mIoU. Similarly, ResNet-101 performs better when Swin-T is its teacher. These results demonstrate that the intrinsic differences between CNNs and transformers are critical for the success of cross-model KD.

We also conducted cross-model distillations between CNN → transformer and transformer → CNN using DSCT to evaluate the effectiveness of the CAFD and TNKD submodules. Table II shows that when Swin-B is selected as the teacher model, the DSCT framework can enhance the student's performance by 1.61% mIoU over the original ResNet-18. Specifically, CAFD and TNKD individually contribute to an improvement of 0.92% and 1.07% mIoU, respectively, for ResNet-18. This result indicates that the attention map and probability distribution can provide complementary and comprehensive knowledge for cross-model distillation. Furthermore, DSCT improves Swin-T's performance by 0.66% mIoU, surpassing that of the teacher model. These findings suggest that DSCT can enable the student model to combine the strengths of the teacher and its own capabilities instead of simply mimicking the teacher.

2) Effectiveness of CAFD: We evaluate the effect of the CAFD module with different loss weight β. In this work, we only apply CAFD for KD, choosing Swin-B and ResNet-18 as the teacher and student models, respectively. The results are presented in Fig. 6. It can be observed that as the loss weight β increases, the performance improvement of the student model ResNet-18 by CAFD presents a trend of first increasing and then decreasing. When β is set to 5, the attention transfer achieves optimal distillation effect. This indicates that CAFD can help the student model focus on the salient features of the teacher model, leading to improved segmentation performance. However, setting the loss weight β too high will cause the student model to rely excessively on imitation rather than exploration.

In addition, normalizations are applied in CAFD to ignore the magnitude information of the attention between the teacher and the student during distillation. We conduct experiments on the ISPRS Potsdam test set to investigate the effects of different normalization dimensions on distillation. As shown in Table III, CAFD achieves better segmentation performance

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TABLE III

ABLATION STUDIES ABOUT THE NORMALIZATION DIMENSIONS OF CAFD. $\bar{H}$, $\bar{W}$, C, AND B DENOTE THE HEIGHT, WIDTH, CHANNEL, AND BATCH OF THE CHANNEL-WEIGHTED FEATURE MAP, RESPECTIVELY. "W/O NORM" MEANS CAFD WITHOUT NORMALIZATION

Method	Norm dimensions	mIoU(%)
T:Swin-B		84.22
S: ResNet-18		82.12
S: ResNet-18 + CAFD	w/o norm	82.31
	$(\bar{H}, \bar{W})$	**83.04**
	$(C, \bar{H}, \bar{W})$	82.97
	$(B, \bar{H}, \bar{W})$	82.88
T: ResNet-101		83.77
S: Swin-T		83.63
S: Swin-T + CAFD	w/o norm	83.74
	$(\bar{H}, \bar{W})$	**84.03**
	$(C, \bar{H}, \bar{W})$	84.02
	$(B, \bar{H}, \bar{W})$	83.96

TABLE IV

ABLATION FOR OUR TNKD MODULE ON THE POTSDAM TEST SET

Method	mIoU(%)
T: Swin-B	84.22
S: ResNet-18	82.12
+ KD [40]	82.95
+ TKD	82.50
+ NKD	82.89
+ TNKD	**83.19**
T: ResNet-101	83.77
S: Swin-T	83.63
+ KD [40]	84.06
+ TKD	83.82
+ NKD	84.01
+ TNKD	**84.12**

TABLE V

PERFORMANCE IN COMPARISON TO OUR TNKD MODULE AT DIFFERENT λ_1 AND λ_2 ON THE POTSDAM TEST SET

λ_1	0.2	0.5	1.0	2.0	5.0
mIoU	83.33	83.45	**83.52**	83.49	83.36
λ_2	1.0	2.0	4.0	8.0	10.0
mIoU	82.97	83.20	83.38	**83.52**	83.48

under different normalization dimensions compared with the case without normalization. Furthermore, the instance normalization along the $(\bar{H}, \bar{W})$ dimension outperforms the layer normalization along the $(C, \bar{H}, \bar{W})$ dimension and the batch normalization along the $(B, \bar{H}, \bar{W})$ dimension. Therefore, we normalize the channel-weighted feature map along the $(\bar{H}, \bar{W})$ dimension when performing CAFD.

3) Effectiveness of TNKD: In this section, we conduct ablations to demonstrate the effectiveness of decoupling the conventional KD into TKD and NKD. As can be seen from Table IV, compared with the conventional KD, the proposed TNKD achieves better segmentation performance. When only NKD is applied, the performances are comparable than the conventional KD. This implies that the knowledge between nontarget classes is crucial for logit distillation, which conveys the relevance of background information. Besides, TKD can also bring about a small performance improvement through soft label distillation. Unlike KD, which only transmits the difficulty information of the target class, TKD also allows the student model to imitate and learn the teacher's decision. Table V reports the mIoU for different values of λ_1 and λ_2, with Swin-B and ResNet-34 set as the teacher and student models, respectively. When one of λ_1 and λ_2 is varied, the other is set to 1. It can be observed that the proposed TNKD module achieves the optimal distillation effect when λ_1 is set to 1.0 and λ_2 is set to 8.0. This indicates that the soft labels from the teacher model can serve as a substitute for ground-truth labels to facilitate student training. In addition, this further underscores the significance of decoupling the weights of TKD and NKD.

D. Comparison to the State-of-the-Arts

In this section, we compare the proposed DSCT method with the state-of-the-art KD methods on four publicly available remote sensing segmentation datasets. We use either ResNet or Swin transformer as the teacher model and train the other model as the student using DSCT.

1) Potsdam: We present a comparison of our proposed DSCT method with five state-of-the-art KD methods (FitNet [48], RKD [44], OFD [41], ReviewKD [49], and DKD [50]) on the Potsdam test set. As demonstrated in Table VI, DSCT enables the compact student model to achieve segmentation performance comparable to that of the teacher model with only 1/3 of the teacher model's parameters. Specifically, the mIoU score improved from 82.25% to 83.73%, and the F_1 score increased from 90.09% to 91.13%. Unlike

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TABLE VI

PERFORMANCE COMPARISON TO FIVE STATE-OF-THE-ART KD METHODS ON THE POTSDAM TEST SET. THE FULL NAMES OF THE CATEGORIES ARE IMPERVIOUS SURFACE, BUILDING, LOW VEGETATION, TREE, AND CAR. THE INDICATOR FOR EACH CATEGORY REPRESENTS THE F_1 SCORE FOR THAT CATEGORY

Method	Imp.Surf.	Building	Low veg	Tree	Car	OA(%)	F_1(%)	mIoU(%)
T:Swin-B	93.74	97.35	87.19	86.81	92.11	91.94	91.44	84.48
S:ResNet-18	92.36	96.52	85.26	85.46	90.83	90.6	90.09	82.25
+ FitNet [48]	92.45	96.64	85.33	85.54	90.82	90.72	90.16	82.62
+ RKD [44]	92.82	96.71	85.37	85.64	90.98	90.88	90.30	82.96
+ OFD [41]	93.17	96.74	86.39	86.04	91.11	91.17	90.69	83.42
+ ReviewKD [49]	93.34	96.86	86.61	86.17	91.48	91.36	90.89	83.59
+ DKD [50]	93.31	96.87	86.68	86.31	91.69	91.42	90.97	83.62
+ DSCT (ours)	**93.47**	**97.06**	**86.84**	**86.42**	**91.87**	**91.54**	**91.13**	**83.73**

TABLE VII

PERFORMANCE COMPARISON TO FIVE STATE-OF-THE-ART KD METHODS ON THE VAIHINGEN TEST SET. THE FULL NAMES OF THE CATEGORIES ARE IMPERVIOUS SURFACE, BUILDING, LOW VEGETATION, TREE, AND CAR. THE INDICATOR FOR EACH CATEGORY REPRESENTS THE F_1 SCORE FOR THAT CATEGORY

Method	Imp.Surf.	Building	Low veg	Tree	Car	OA(%)	F_1(%)	mIoU(%)
T:ResNet-101	87.15	91.89	73.69	85.03	76.00	84.75	82.75	71.16
S:Swin-T	86.36	90.05	72.28	84.66	67.53	83.65	80.18	67.78
+ FitNet [48]	86.29	90.14	72.24	84.57	68.02	83.66	80.25	67.80
+ RKD [44]	86.47	90.62	72.58	84.71	68.52	83.82	80.58	67.91
+ OFD [41]	86.52	90.57	72.61	84.62	68.89	83.81	80.64	67.99
+ ReviewKD [49]	86.86	**91.21**	73.04	85.14	67.59	84.02	80.77	68.52
+ DKD [50]	86.54	91.02	73.13	84.72	**73.17**	84.32	81.72	69.25
+ DSCT (ours)	**87.26**	91.09	**74.87**	**85.65**	72.60	**84.90**	**82.29**	**70.55**

the other five KD methods that solely rely on feature distillation or logit distillation, our proposed DSCT combines both the paradigms to deliver better segmentation performance improvements. In addition, DSCT does not introduce any extra training parameters and can be trained end-to-end. In Fig. 7, the upper three rows illustrate several visualized results on the Potsdam test set, which further demonstrate the effectiveness of our proposed method.

2) Vaihingen: The comparison results for the Vaihingen test set are presented in Table VII. The Vaihingen dataset contains only 142 training samples after nonoverlapping cropping, which poses a significant challenge for transformers, as they require a large amount of labeled data for fitting. Given that the compact student model Swin-T is incapable of delivering satisfactory segmentation performance, the proposed DSCT can enhance the mIoU of the student by 2.77 points, resulting in similar accuracy to the cumbersome teacher model. Furthermore, all the KD methods can enhance the student's performance under the teacher's supervision, and our DSCT attains the best segmentation performance. Specifically, for the category Car, our method achieves 5.07% IoU improvement, which is much higher than feature-based KD methods such as OFD. This indicates that logit probability alignment can help students achieve better segmentation performance than feature matching when training samples are small. From the bottom three rows of Fig. 7, it is evident that our DSCT generates more consistent and detail-preserving predictions than ReviewKD, which confirms the effectiveness of our method.

3) GID: As shown in Fig. 8, we present the F_1 score for each individual class of the student model ResNet-34 when Swin-B is selected as the teacher model. Our DSCT method outperforms the baseline (w/o distillation) and DKD, achieving better class F_1 scores, particularly for categories with lower F_1 scores. For instance, our DSCT method demonstrates a 5.45% and 2.34% relative improvement on Pond compared with the baseline and DKD, respectively. In addition, Fig. 9 visually presents the qualitative segmentation results, which further illustrate the effectiveness of our proposed method.

4) LoveDA: We report the results and the comparisons to the state-of-the-art semantic segmentation models and KD methods on the LoveDA urban validation set. As shown in Table VIII, the proposed DSCT helps the compact semantic segmentation model achieve significant improvements. Specifically, the mIoU is improved from 48.12% to 50.89%. Besides, the compact ResNet-18 outperforms DeepLabV3+ and PSPNet with fewer parameters. Fig. 10 shows several

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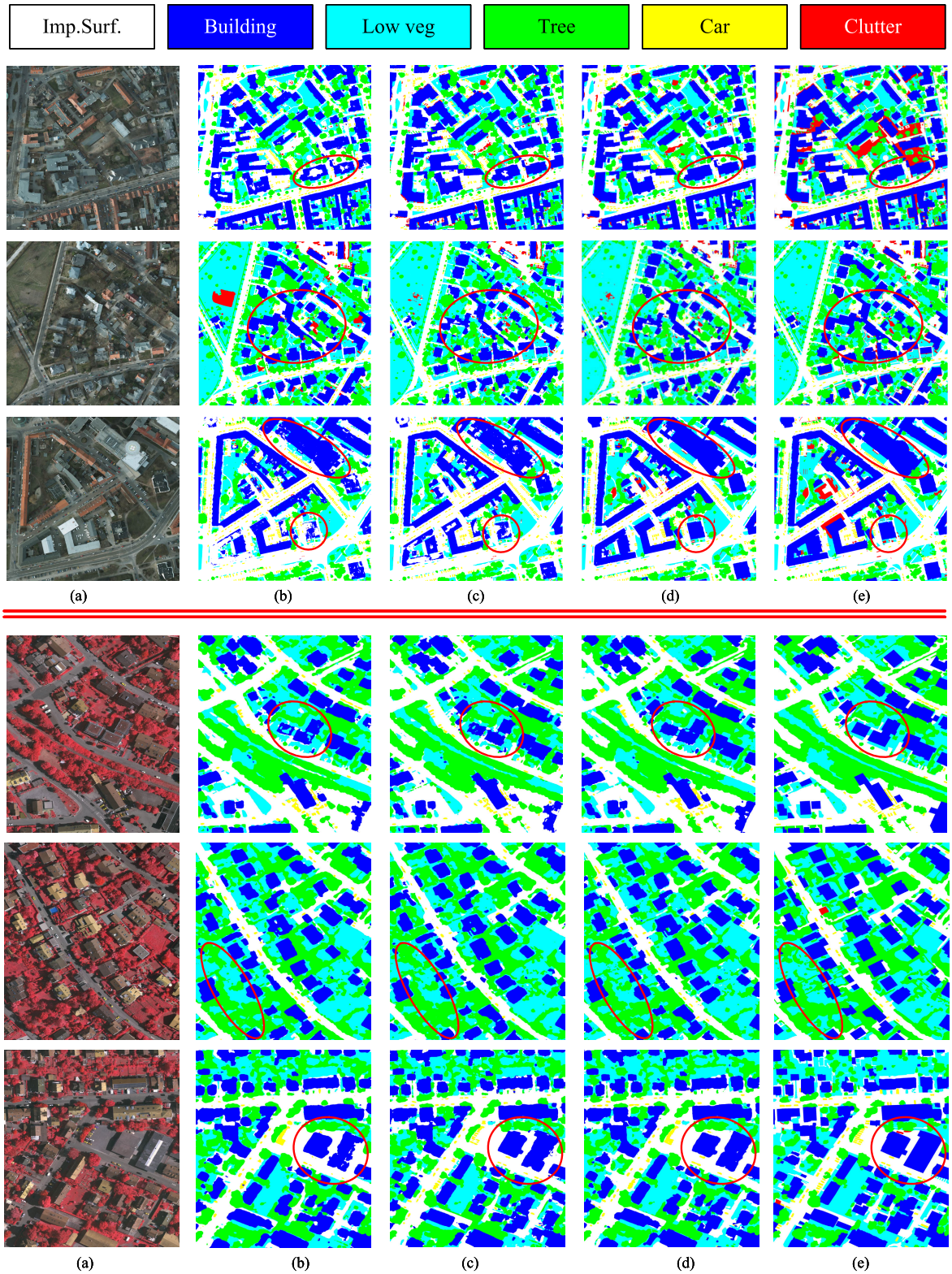


Fig. 7. Performance comparison between different KD methods on the Potsdam test set (the upper three rows) and Vaihingen test set (the bottom three rows). "w/o dis." represents the baseline student model trained without KD. (a) Input image. (b) w/o dis. (c) Results of OFD (the upper three rows) and results of ReviewKD (the bottom three rows). (d) Results of ours. (e) Ground truth.

samples on the LoveDA urban test set, and better segmentation results are achieved by the proposed DSCT, which also further demonstrate the effectiveness of our method.

To demonstrate the generalization and robustness of our proposed method, we also conduct transfer learning experiments on the LoveDA dataset. In the experiments, we use the urban part of the LoveDA dataset as the source domain and the rural part as the target domain. We first pretrain a cumbersome teacher model (Swin-B) on the source domain. Using the source domain knowledge and the target source

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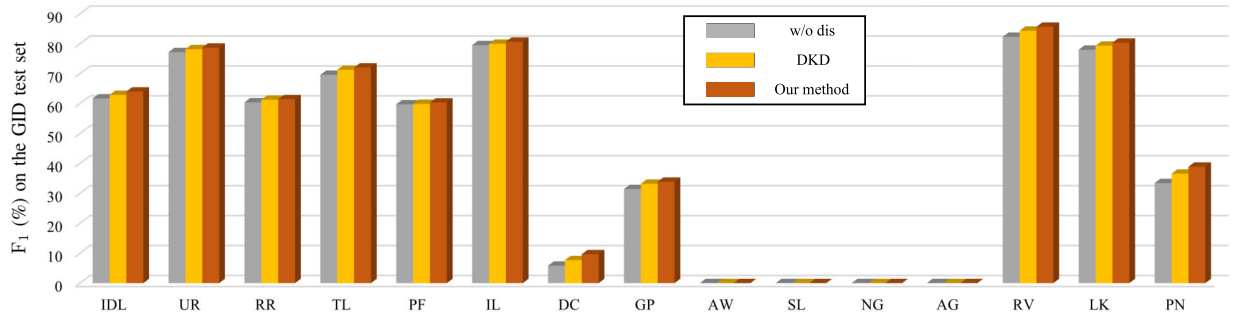


Fig. 8. Illustration of individual class F_1 scores over the student model ResNet-34 with baseline (without distillation), DKD, and our proposed DSCT on the GID test set. The full names of the categories are industrial (IDL), urban resident (UR), rural residential (RR), traffic land (TL), paddy field (PF), irrigated land (IL), dry cropland (DC), garden plot (GP), arbor woodland (AW), shrub land (SL), natural grassland (NG), artificial grassland (AG), river (RV), lake (LK), and pond (PN). "w/o" represents the student model is trained without distillation.

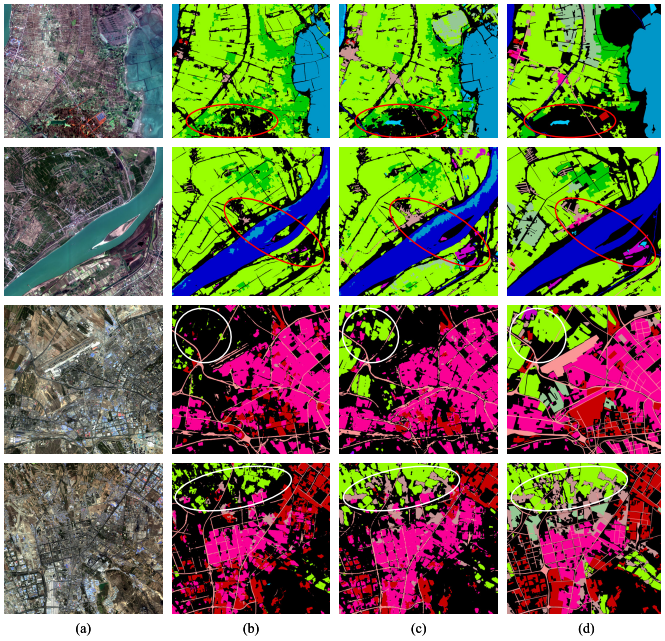


Fig. 9. Qualitative segmentation result comparisons on the GID test set. The proposed DSCT framework improves the performance of the student model. (a) Input images. (b) Results of the student model. (c) Results of ours. (d) Ground truth.

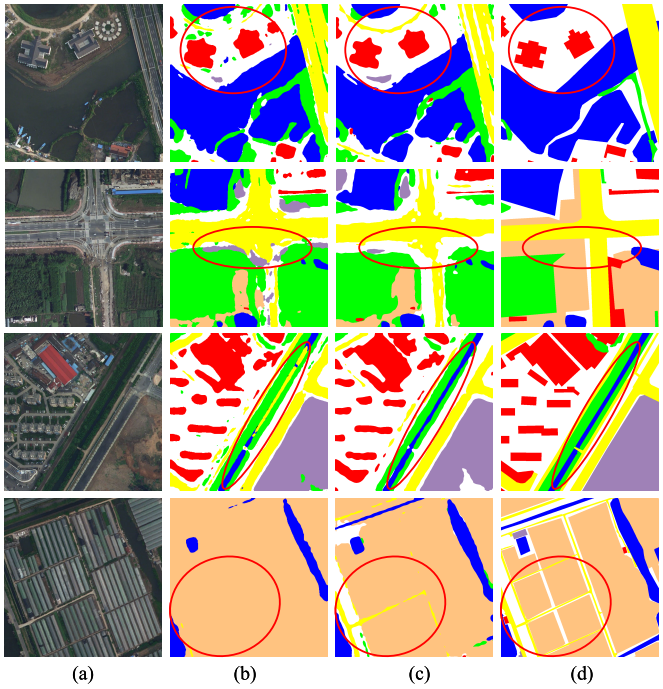


Fig. 10. Visualizations of the results of different methods on the LoveDA urban validation set. (a) Input images. (b) Results of the student model. (c) Results of ours. (d) Ground truth.

TABLE VIII
PERFORMANCE COMPARISON ON THE LOVEDA URBAN VALIDATION SET

Method	Params.	OA(%)	F_1(%)	mIoU(%)
DeepLabV3+ [68]	62.57	63.71	65.94	50.03
PSPNet [13]	67.96	63.86	66.18	50.44
SETR [20]	309.57	68.06	68.97	53.62
T:Swin-B	121.28	67.62	68.39	53.04
S:ResNet-18	41.50	61.13	63.85	48.12
+ FitNet [48]	41.50	61.69	64.08	48.77
+ ReviewKD [49]	41.50	63.58	65.89	50.27
+ DSCT(ours)	41.50	**64.29**	**66.72**	**50.89**

TABLE IX
DOMAIN ADAPTATION RESULTS OBTAINED ON THE LOVEDA DATASET

Method	OA(%)	F_1(%)	mIoU(%)
No Adaptation	66.07	53.07	38.02
Domain Adaptation	**69.81**	**58.14**	**44.73**

dataset, a compact student model (ResNet-18) is then trained via our proposed DSCT framework. As seen in Table IX, compared with the no adaptation method, the proposed DSCT method shows significant improvements. This suggests that our method is able to address the problems of spatial resolution inconsistency and class distribution and spectral differences, thereby improving the model transferability in real remote sensing scenarios.

V. CONCLUSION

In this article, a novel cross-model KD framework is proposed, called DSCTs, which aims to enhance the segmentation performance of the CNN-based and transformer-based segmenters by distilling complementary strengths from each other. Specifically, a CAFD module is introduced, which leverages the spatial attention maps to learn the channelwise

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aggregated activations and spatial dependencies. Furthermore, a TNKD module is designed, which decouples the categorical logits to enhance flexibility and unsuppress the effect of nontarget knowledge. The ablation experiments demonstrate the potential of the proposed cross-model distillation approach in semantic segmentation tasks, as well as the efficacy and robustness of our proposed framework. In addition, we conduct comparison experiments on four publicly available remote sensing semantic segmentation datasets, and the experimental results demonstrate the effectiveness of our approach, as well as its superior performance compared with the state-of-the-art distillation methods. As a further development of this article, we plan to investigate the interpretability of KD and consider applying our method to other dense prediction tasks.

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Zhe Dong received the M.S. degree in software engineering from Harbin Engineering University, Harbin, China, in 2022. He is currently pursuing the Ph.D. degree with the School of Electronics and Information Engineering, Harbin Institute of Technology (HIT), Harbin.

His research interests include semantic segmentation and self-supervised learning of remote sensing images.



Guoming Gao (Member, IEEE) received the Ph.D. degree in information and communication engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2018.

He joined as a Research Assistant with the School of Electronics and Information Engineering, HIT. His research interests include multimodal image analysis, remote sensing image classification and target detection, and missing data recovery.



Tianzhu Liu (Member, IEEE) received the Ph.D. degree in information and communication engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2019.

She was a Lecturer with the School of Electronics and Information Engineering, HIT, where she is currently an Associate Professor and a Post-Doctoral Fellow. Her research interests include image processing in hyperspectral remote sensing, especially multimodal hyperspectral image classification.

Dr. Liu won the Third Prize in Student Paper Contest at the International Geoscience and Remote Sensing Symposium (IGARSS) in 2019. She serves as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING and a reviewer for several international journals, such as IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and *Neurocomputing*.



Yanfeng Gu (Senior Member, IEEE) received the Ph.D. degree in information and communication engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2005.

He joined as a Lecturer with the School of Electronics and Information Engineering, Harbin Institute of Technology (HIT). He was appointed as an Associate Professor at HIT in 2006; meanwhile, he was enrolled in first Outstanding Young Teacher Training Program of HIT. From 2011 to 2012, he was a Visiting Scholar with the Department of Electrical Engineering and Computer Science, University of California, Berkeley, Berkeley, CA, USA. He is currently a Professor with the Department of Information Engineering, HIT. He has published more than 100 peer-reviewed papers, four book chapters, and is the inventor or coinventor of 20 patents. His research interests include space intelligent remote sensing and information processing, multimodal hyperspectral remote sensing, and spaceborne time-series image processing.



Xiangrong Zhang received the Ph.D. degree in management science and engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2016.

She was a Lecturer with the School of Management, Heilongjiang Institute of Technology, Harbin, where she is currently an Associate Professor. Her research interests include remote sensing data analysis and time-series prediction.

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